

Shruti A. Panicker

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Seeking full-time opportunities from July 2026 | Open to remote

SUMMARY

Data Scientist and ML practitioner with over 4.5 years of experience across product analytics, experimentation, and applied machine learning, translating behavioral and business data into product and engineering decisions for Fortune 20 clients. Combines industry experience in decision science with graduate research in NLP, LLM evaluation, and voice AI systems.

EDUCATION

M.S. in Artificial Intelligence – ML Concentration, *Northeastern University* Boston, MA, USA | Sep 2024 – May 2026

GPA: 3.875/4 | **Coursework:** Foundations of AI, Data Processing for AI, Machine Learning, NLP, Human Computer Interaction

B.Tech in Computer Science and Engineering, *Amrita Vishwa Vidyapeetham* Coimbatore, India | Jul 2016 – Aug 2020

GPA: 8.06/10 | **Coursework:** Data Structures, Computational Intelligence, Deep Learning, Data Science

EXPERIENCE

Graduate Teaching Assistant, *Northeastern University* Boston, MA, USA | Sep 2024 – Apr 2026

- Led instruction for DS2500 across all 4 MS semesters (250+ students/semester), redesigning assignments and project structure.

ML Engineer, *NexeraLabs Pvt Ltd* Trivandrum, India | Jun 2024 – Aug 2024

- Built an active-learning scoring pipeline that predicted moderator rubric scores from diarized transcript features (turn-taking, talk-time ratio, topic-keyword coverage), routing only low-confidence sessions for human review, **reduced moderator review load by 65%** while maintaining scoring accuracy within 3 points of moderator consensus.

Data Scientist Lead, *Mu Sigma Business Solutions* Bangalore, India | Jan 2024 – May 2024

- Directed a 10-person decision science team delivering SaaS product analytics for a Fortune 20 technology client's global collaboration device portfolio, translating analysis into roadmap decisions presented to C-suite leadership.
- Built a statistical upsell propensity model in Python and SQL, applying TAM/SAM/SOM segmentation across large-scale datasets to identify high-conversion targets and **drive 5% product adoption growth** through focused campaigns.
- Set fiscal year targets for 5+ OKRs using time series forecasting, establishing a statistical target-setting framework where none existed.
- Translated analytical insights into product strategy and investment decisions by partnering with product and business leadership on C-suite presentations, directly influencing roadmap priorities.

Decision Scientist, *Mu Sigma Business Solutions* Bangalore, India | Sep 2020 – Dec 2023

- Built a graph-based entity resolution system to deduplicate Device IDs across large-scale datasets, applying network association analysis to identify anomalous identity patterns and restore integrity across all downstream portfolio metrics.
- Designed a three-state customer lifecycle model (inactive/churned/reactivated) using classification and behavioral segmentation, enabling targeted re-engagement campaigns that **drove 18% engagement increase within 6 months**.
- Diagnosed product onboarding drop-off by joining behavioral usage signals with support data in Python and SQL, pinpointing friction points across onboarding stages and recommending targeted interventions that **drove a 30% increase in reactivation rates**.
- Led 8-month EUDB-compliant pipeline migration across 25+ dashboards, redesigning ETL transformation and aggregation layers to ensure EU data residency compliance with zero loss in metric accuracy.

Data Analyst, *Riveron Infotech Pvt Ltd* Trivandrum, India | Jan 2020 – Aug 2020

- Conducted cohort analysis tracking student progression across program levels, identifying drop-off points that drove targeted interventions and **improved program completion rates by 18%**.
- Built and analyzed a 6-stage website-to-paid-enrollment funnel in Python, SQL, and Tableau across 5K+ sessions, identifying highest-friction drop-off points and recommending UX improvements projected to **lift paid enrollment conversion by 15%**.
- Built a churn prediction model using behavioral signals (login frequency, activity completion, mentor engagement) to flag at-risk students, enabling proactive outreach that **improved retention by 30%**.

PROJECTS

Semantic Turn Detection for Voice AI *Northeastern University*

Built an intent-conditioned voice-agent turn-detection system on a Groq-hosted Llama-3.3-70B ensemble, classifying utterances as transactional, informational, or reflective and dynamically adapting response timing per intent type. Developed annotation guidelines and a within-subjects evaluation (n=18); versus a silence-only VAD baseline, Naturalness rose 4.31→6.00, Frustration dropped 3.08→1.53 on 7-point Likert scales.

Empathy Detection – WASSA 2024 Shared Task *Northeastern University*

Fine-tuned BERT for regression across 11K+ conversational data points to predict empathy, emotional polarity, and intensity. Designed intrinsic and extrinsic evaluation frameworks to validate model behavior, achieving 0.597 average Pearson correlation, which is a score that would have placed in the top 4 on the WASSA 2024 shared task leaderboard.

SKILLS

Languages & Query: Python, SQL, R, C#, SCOPE, Power BI DAX, Git/GitHub | **ML & Modeling:** Pandas, NumPy, PyTorch, TensorFlow, Keras, scikit-learn, BERT/Transformers, Hugging Face, Statistical Modeling | **NLP & LLMs:** NLP, LLM Evaluation, Prompt Engineering, Responsible AI | **Data & Analytics:** A/B Testing, User Behavior Analysis, Cohort & Funnel Analysis, Time Series Forecasting | **Data Engineering:** Apache Spark, PySpark, AWS, Azure, Hadoop | **Visualization & BI:** Power BI, Tableau | **Certifications:** AWS Certified Cloud Practitioner (ID - Y9C4WRJCBJVEQ1KL), AWS Certified Machine Learning Engineer Associate (ID - 8a34b25eb698433dab2ceef1ff2c0ccb)